

Football Match Manager - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager API provides endpoints for managing football matches, including registration, login, and logout. It also includes endpoints for retrieving matches, teams, and leagues, as well as following and unfollowing matches. The API uses JWT tokens for authentication and has rate limiting in place.

1.2 Maturity Assessment

The API specification is mostly complete, but there are some gaps in the documentation, including missing sections for goals and objectives, functional requirements, and non-functional requirements. The API has a clear structure and uses standard HTTP status codes for error handling. However, the rate limiting and authentication mechanisms may need to be refined. Overall, the API is in the REFINEMENT stage, with some work needed to complete the documentation and refine the implementation.

1.3 Technical Stack

- Python
- Flask
- JWT

1.4 Architectural Constraints

- Rate limiting: 5 requests per minute per IP for auth endpoints, 60 requests per minute per user for all other endpoints
- Authentication: JWT tokens required for all endpoints except auth endpoints
- Versioning: API version included in URL path (/api/v1/)
- Error handling: HTTP status codes (200, 201, 400, 401, 403, 404, 429, 500) with error response body

1.5 Critical Dependencies

- JWT tokens for authentication
- Database for storing match, team, and league data
- Rate limiting mechanism to prevent abuse
- Error handling mechanism to handle unexpected errors

2. Architecture Workflows & Visual Diagrams

The API's architecture can be visualized through the following diagrams:

- Entity Relationship Diagram: illustrates the relationships between entities in the Football Match Manager API
- Sequence Diagram for Authentication: shows the sequence of events for user authentication

- Flowchart for Matches Endpoint: outlines the flow of the matches endpoint
- Flowchart for Rate Limiting: describes the rate limiting mechanism

3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Description	Source Section
MATCH-01	Match entity	Match
TEAM-01	Team entity	Team
LEAGUE-01	League entity	League
USER-01	User entity	User (authenticated user only)
AUTH-01	Authentication requirement	Authentication
MATCHES-01	Matches endpoint	Matches
FOLLOWS-01	Follows endpoint	User Follows
RATE-LIMITING-01	Rate limiting requirement	Rate Limiting
VERSIONING-01	Versioning requirement	Versioning

3.2 Security Rules

The API uses JWT tokens for authentication and has rate limiting in place to prevent abuse.

3.3 Data Models

The API uses the following data models:

- Match: represents a football match
- Team: represents a football team
- League: represents a football league
- User: represents an authenticated user
- Follow Response: represents a user's follow response

4. Project Governance & Structural Gaps

4.1 Structural Gaps

The following sections are missing from the API documentation:

- Goals & Objectives
- Functional Requirements
- Non-Functional Requirements

- Scope & Out-of-Scope
- Open Questions & Uncertainties

4.2 Remediation & Workflow

To address these gaps, the following remediation steps are recommended:

- Add a section to describe the goals and objectives of the API
- Add a section to describe the functional requirements of the API
- Add a section to describe the non-functional requirements of the API
- Add a section to describe the scope and out-of-scope items for the API
- Add a section to describe any open questions or uncertainties related to the API

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	API Contract: Football Match Manager	The primary interface for interacting with the Football Match Manager backend, providing endpoints for various operations.
CORS Standard	TECHNICAL_STACK	API Contract: Football Match Manager	A set of rules governing cross-origin resource sharing, ensuring secure data exchange between web pages and servers.
ID	BUSINESS_DOMAIN	MATCH-01	A unique identifier assigned to each entity, such as a match, team, or user, for tracking and reference purposes.
IP	TECHNICAL_STACK	Rate Limiting	A numerical label assigned to each device connected to a network, used for communication and identification.
JSON	TECHNICAL_STACK	Data Models	A lightweight data interchange format, used for exchanging data between servers, web applications, and mobile apps.
JWT	TECHNICAL_STACK	Authentication	An open standard for securely transmitting information between parties, used for authentication and authorization.

UUID	TECHNICAL_STACK	MATCH-01	A 128-bit label used for information in computer systems, providing a unique identifier for each entity.
dateFrom	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the start date for the range of matches to be retrieved.
dateTo	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the end date for the range of matches to be retrieved.
leagueId	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the league for which matches are to be retrieved.